

Cross-Asset Portfolio Construction as Cashflow Engineering

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What This Paper Decides

This study is a capital-admission audit for a SPY-centered investor. Each candidate sleeve is translated into funded cashflows, charged for friction, and tested under a downside objective before it can receive risk budget. The answer is deliberately selective: the cross-asset downside book is admitted, option sleeves are qualified as state-contingent defense, and machine-learning forecasts remain research inputs rather than headline trades.

Abstract

This paper studies cross-asset allocation as an admission decision. Starting from SPY, it asks whether a constituent equity basket, gold, copper, VIX futures, Treasuries, and SPY options deserve capital after funding, transaction costs, option premia, drawdowns, and tail losses are charged. Each sleeve is specified by payoff, mark, margin, hedgeability, and friction before it is allowed to compete with the benchmark.

The empirical design builds a daily execution panel and aggregates executed P&L to a monthly evaluation horizon. A rolling out-of-sample test from February 2018 through May 2026 is checked with purged cross-validation, backtest-overfitting diagnostics, block bootstrap, volatility-clustered Monte Carlo, cost multipliers, and a gated forecast arm. The main results are simple: the no-option cross-asset downside book is the strongest headline allocation, the option-enabled book is more defensive but gives up return, and the ML arm does not clear the same portfolio-level admission bar.

Key Findings

- Admitted: the no-option cross-asset downside book.
- Qualified: options as state-contingent defense, not return enhancement.
- Rejected for headline use: ML forecasts that do not beat the simpler rule.

Impact Statement

Cross-asset research often compares return columns before specifying the cashflows that produce them. This paper reverses the order: each sleeve states its payoff, mark, funding, hedge, data-quality, and friction conventions before optimization. The mixed evidence is the decision. Diversification survives validation, option convexity is qualified by state and cost, and machine-learning forecasts stay out of the headline allocation until they improve the whole portfolio.

1. Introduction

An investor who already owns SPY has a simple question with a hard implementation: which additional sleeves deserve capital once their cashflows, costs, and downside effects are made explicit? Adding more columns to a return matrix can almost always improve an in-sample frontier. The more useful test is whether a sleeve survives after it is funded, hedged, rolled, charged for trading costs, and evaluated out of sample.

This paper treats the proposed universe as a capital-admission audit. The running book contains SPY, a liquid large-cap constituent basket, gold, copper, VIX futures, Treasuries, and SPY option overlays. These are not interchangeable return labels. VIX futures can pay during volatility shocks but are expensive to hold through roll. Treasuries can hedge equity weakness in some regimes and fail in inflation shocks. Options create convexity, but the investor pays premium, skew, spread, theta, and sometimes hedge turnover before any crash protection is earned.

1.1. *What the paper decides*

The admission question is whether a SPY-centered investor should allocate risk budget to a sleeve after it has been translated into funded, executable cashflows and evaluated under downside objectives. The allocation problem is therefore not to find the best standalone return column. It is to decide whether an instrument improves the total book after funding, roll, premium, spread, hedge cost, turnover, drawdown, and left-tail loss are all charged.

The paper reports three decisions. First, the no-option cross-asset downside book is admitted as the headline allocation in this sample. It improves the tested SPY path and survives the main validation screens. Second, option sleeves are qualified rather than promoted unconditionally. They improve several downside and tail measures when admitted inside the allocator, but they give up return relative to the no-option book. Third, machine-learning forecasts are rejected for headline allocation. They remain documented research inputs because fold-level forecast evidence does not translate into a better portfolio than the simpler downside rule.

The non-claims are equally important. The evidence supports a disciplined research workflow, not live trading performance. The study does not claim brokerage-level execution quality, production-grade alpha, or an unconditional reason to buy option overlays. A production result would require richer market data, executable roll and option quotes, larger point-in-time universes, and live cost measurement.

1.2. *Positioning*

Mean-variance portfolio theory frames allocation as a tradeoff between expected return and covariance risk [1]. That framework remains the reference point, but this mandate is more asymmetric than classical mean-variance selection. Downside risk measures such as Sortino ratio, lower partial moments, coherent tail risk, and CVaR optimization better match an investor who cares about drawdowns and left-tail capital impairment [2, 4, 3].

Robust portfolio optimization responds to unstable estimates by penalizing uncertain means and covariances or by optimizing across ambiguity sets [16, 17]. Shrinkage and disciplined diversification offer practical responses when the sample is short relative to the instrument menu [15, 14]. The implementation below uses these ideas as admission hurdles, not as proof that estimated expected returns are reliable trading signals.

Option pricing and overlay design add the central qualification. Listed options can supply convexity, but option-pricing and transaction-cost replication results emphasize that convexity is paid for through premium, spread, skew, theta, and hedge turnover [5, 6, 7, 12]. A protective put is therefore not evaluated here as a free drawdown reducer. It must pay in the portfolio states where the book needs convexity, and it must do so after option marks and hedge costs are recorded.

Backtest validation supplies the final gate. Financial machine-learning and strategy selection work stresses purging, embargo, and probability-of-backtest-overfitting checks when many candidate rules compete over a short realized history [8, 13]. The validation layer follows that warning by keeping negative evidence visible. Forecasts and overlays are admitted only after they survive portfolio-level tests, not because they improve a standalone score.

The remainder follows the audit workflow. Section 2 defines the data contract. Section 3 builds the cashflow framework. Section 4 states the admission rule and portfolio objective. Section 5 describes the validation gates. Section 6 reports the admitted, qualified, and rejected evidence. Section 7 concludes with limits and production gaps. Appendix A preserves the full diagnostics and implementation details.

2. Data

The empirical design starts with the investment universe and the measurement frequency. The benchmark is SPY. Candidate sleeves are a liquid large-cap stock basket, gold futures, copper futures, a tradable VIX-futures proxy, TLT as the rates sleeve, and SPY option overlays. The sample runs from February 2015 through May 2026 after excluding incomplete terminal months, and the reported rolling experiment begins in February 2018 after a 36-month training window.

The allocation frequency is monthly because the research question is strategic portfolio construction rather than intraday trading. The execution frequency is daily because option marks, futures rolls, hedge trades, transaction costs, and drawdown control are not naturally monthly objects. At each monthly review date the allocator sees only completed historical data. The selected target is then passed to a daily execution layer that records net P&L, trading costs, option lifecycle events, hedge actions, futures rolls, and drawdown-control records before returns are aggregated back to the monthly evaluation horizon.

Every reported number must be traceable to a dated input panel, and every instrument must enter through a documented convention. The base run uses processed research panels for returns, option marks, futures proxies, transaction costs, and validation folds. Tables and figures are rebuilt from those inputs rather than edited by hand.

Object	Implementation
Sample	2015-02-28 to 2026-05-31
Core observations	136
Expected calendar months	136
Missing core month-ends	0
Option overlay months	127
Equity index	SPY from local adjusted-close cache
Stock subset	AAPL, MSFT, NVDA, AMZN, GOOGL, JPM, XOM, UNH equal-weight return
Gold/copper	GC=F and HG=F local continuous futures
Volatility	Nearest VIX future proxy, not spot VIX
Rates	TLT proxy; ZN=F retained in panel
Options	SPY nearest-listed 95% put / 105% call and 95/105 collar targets

Table 1: Data objects used by the SPY-centered empirical experiment. The table records the sample, major sleeves, option sources, and proxy conventions used in the analysis.

The universe is intentionally narrow enough to audit and broad enough to exercise the framework. The stock sleeve is an equal-weight basket of liquid large-cap constituents. Gold and copper use local continuous futures proxies. VIX exposure is represented by a tradable VIX-futures proxy rather than spot VIX. Rates enter through TLT, with rates futures retained as diagnostics. SPY options are selected from month-end OPRA slices as nearest-listed 95% put and 105% call targets. These choices keep the experiment tied to instruments a portfolio could plausibly trade while avoiding a claim that the public research panel captures every venue-level execution detail.

Instrument	Class	Accounting	Main friction
SPY	equity ETF	paid spot exposure plus distributions	spread, impact, cash financing
EQ_BASKET	large-cap stocks	paid spot basket return	survivorship, corporate actions, single-name liquidity
GC_FUT	gold future proxy	variation-margin return from continuous proxy	roll, margin, delivery-calendar simplification
HG_FUT	copper future proxy	variation-margin return from continuous proxy	roll, margin, cyclicalities
VX_FRONT	VIX future proxy	front-contract futures return	roll drag, convex stress behavior, non-spot tradability
TLT	Treasury ETF proxy	paid duration exposure	DV01, financing, rates-regime risk
SPY_95P	listed option overlay	premium paid, expiry payoff or sale/exercise value	bid/ask, skew, theta, quote timing
SPY_95_105_COLLAR	listed option overlay	long put plus short call terminal cashflow	assignment, capped upside, bid/ask

Table 2: Instrument normalization contract. Each sleeve is represented by its accounting object and the main friction that must be charged before the allocator can use it.

The data are research-quality, not execution-complete. Continuous futures are proxy series. The VIX-futures roll rule is deterministic rather than a full venue-level execution engine. SPY option overlays use quote anchors and daily model marks when daily matched OPRA quotes are unavailable. The stock basket is representative rather than a full point-in-time S&P 500 reconstruction. Public and reproducible data constraints therefore bound the empirical claim. They are acceptable for a working paper that tests whether a cashflow-accounting framework behaves sensibly across sleeves, but they are not sufficient for a live trading claim or a production execution study.

3. Cashflow and Instrument Framework

The central modeling step is to express every sleeve as a domestic-currency cashflow. Let ϕ_i denote the convention for instrument i : payoff map, mark, currency, funding account, margin or premium rule, exposure vector, and friction profile. A portfolio x has scenario cashflow

$$\Pi_s(x) = \sum_i x_i \pi_{i,s}^{\phi_i} - C_s^\phi(x),$$

where C_s^ϕ includes spreads, commissions, impact, financing, margin drag, roll cost, hedge cost, and data-quality penalties. The object optimized later is not a raw asset return. It is this friction-deflated, convention-specific cashflow.

Instrument	Linear exposure	Convexity	Vol/rates exposure	Data risk	Funding
SPY	equity beta	none	none	low	paid cash
EQ_BASKET	single-name beta	none	none	medium	paid cash
GC_FUT	commodity delta	none	none	medium	variation margin
HG_FUT	commodity delta	none	none	medium	variation margin
VX_FRONT	volatility-futures delta	roll convexity	vol regime	high	variation margin
TLT	duration beta	bond convexity	key-rate proxy	medium	paid cash
SPY_95P	negative equity delta	gamma	vega/theta	high	premium
SPY_95_105_COLLAR	reduced equity delta	put/call gamma	skew/theta	high	premium net short-call risk

Table 3: Exposure map used by the allocator. The table separates linear exposure, convexity, volatility or rates sensitivity, data risk, and funding convention.

Equities and ETFs are paid spot exposures with dividends, borrow, tracking, and liquidity considerations. Futures are variation-margin instruments, so their wealth effect is not the same as buying the underlying with cash. FX forwards combine spot movement, domestic and foreign funding, forward points, and roll spread. Bonds are coupon strips with duration, key-rate, convexity, and financing exposure. Options require premium, exercise, settlement, volatility-surface, and hedge conventions before their payoff can be compared with a simple equity sleeve.

The option convention is particularly important. A put bought at time t for premium $P_t(K, T)$ contributes

$$R_{t,T}^{put} = \frac{(K - S_T)_+ - P_t(K, T)}{S_t}.$$

This is a realized physical-measure contribution after paying the market premium. The collar adds the short-call leg:

$$R_{t,T}^{collar} = R_{t,T}^{put} + \frac{C_t(K_c, T) - (S_T - K_c)_+}{S_t}.$$

The formulas explain why options can reduce drawdown and still reduce average return. The premium is paid in every state, while the convex payoff arrives only in selected states.

Hedging is modeled as a policy, not as a free continuous replication assumption. Option

delta and gamma enter the exposure map, and the daily execution layer applies no-trade bands before hedge trades are recorded. Rates and futures exposures similarly require roll and key-rate conventions. A sleeve that appears attractive before these details are charged is not yet an allocation candidate.

4. Portfolio Construction Model

The portfolio model is an admission rule, not a tournament of optimizers. A sleeve enters only when its net portfolio contribution is positive after funding, spread, roll, premium, hedge cost, turnover, drawdown, and tail risk are charged. That rule lets different instruments compete on the same economic object: domestic cashflow delivered to the SPY-centered book.

The primary criterion is robust downside contribution. For target return τ , portfolio return $R(x)$, and ambiguity set $\mathcal{U}$, the guiding objective is

$$\max_x \inf_{u \in \mathcal{U}} \frac{\mathbb{E}_u[R(x)] - \tau}{\sqrt{\mathbb{E}_u[(\tau - R(x))_+^2]}}$$

subject to budget, gross, liquidity, margin, stress, and turnover constraints. The denominator is the square root of a lower partial moment. Upside volatility does not penalize the book in the same way as losses below the target.

Tail and path constraints sit beside Sortino. CVaR controls the average loss beyond a quantile:

$$\text{CVaR}_\alpha(L) = \min_\eta \left\{ \eta + \frac{1}{1 - \alpha} \mathbb{E}[(L - \eta)_+] \right\}.$$

Drawdown is pathwise. A hedge can be valuable even with negative standalone mean if it pays in the states that set the running peak-to-trough loss. This is the economic reason options and VIX futures are evaluated by portfolio contribution rather than by standalone average return.

The public reproduction path uses deterministic candidate search with caps, turnover limits, friction charges, and concentration penalties. That keeps the code auditable while preserving the intended economics of robust MVO, Sortino, CVaR, and drawdown-aware allocation. The full optimizer menu is left to Appendix A because the body decision does not depend on selling a new optimizer.

In words, the admission rule is:

admit sleeve j only if $\Delta \text{portfolio objective}_j$ remains positive after costs and validation.

Equities must add compensated beta or factor exposure. Commodities must add useful inflation, growth, or real-asset state exposure. VIX futures must overcome roll drag with stress payoff. Treasuries must provide duration or key-rate payoff in the relevant regime. Options must provide convex payoff net of premium, skew, spread, theta, and hedge cost.

5. Metrics and Validation

The validation design asks four questions before a sleeve can be admitted. The realized path asks what happened from February 2018 through May 2026 after daily execution was aggregated to monthly returns. CPCV asks whether the fitted rule depends on a particular ordering

of held-out months. Simulation and drawdown-breach checks ask whether the same strategy revisits dangerous loss regions under alternative path assumptions. Cost stress asks whether a result disappears when frictions are less favorable.

The body keeps the validation layer decision-readable. The two path figures show the realized wealth and drawdown evidence. The compact validation table in Section 6 reports the gates that decide whether evidence is admitted, qualified, or rejected. Full CPCV, PBO, simulation, stress, execution, option-state, and ML tables are preserved in the appendix for auditability.

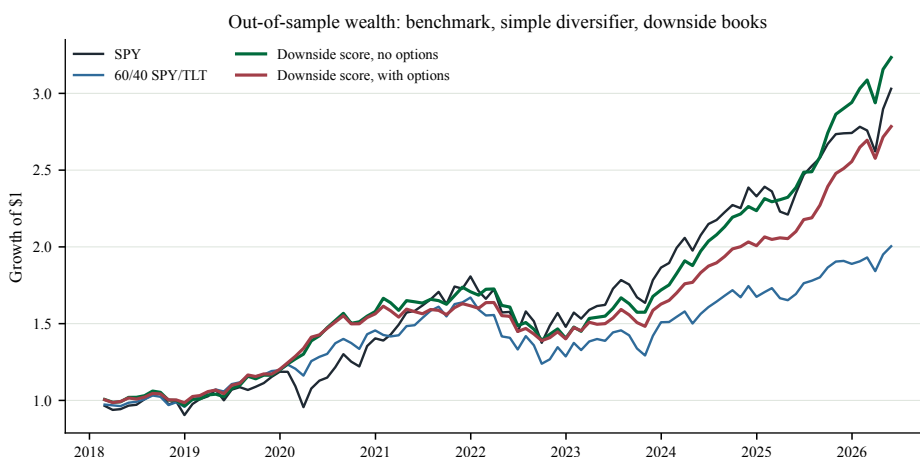


Figure 1: Out-of-sample wealth paths from February 2018 through May 2026. The display asks what survives after daily execution is aggregated to monthly returns: SPY remains the benchmark, the balanced sleeve is the simple diversifier, and the downside-score lines show whether the cashflow-aware allocator adds value with and without option convexity.

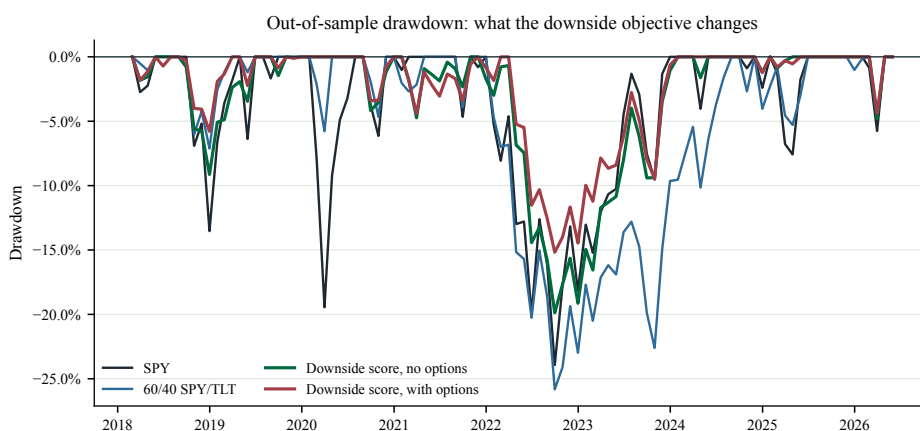


Figure 2: Out-of-sample drawdown paths from February 2018 through May 2026. The figure shows why the result is a downside-risk claim rather than a return-only claim: the reader should compare depth, duration, and recovery across the benchmark, simple diversifier, and two downside-score books.

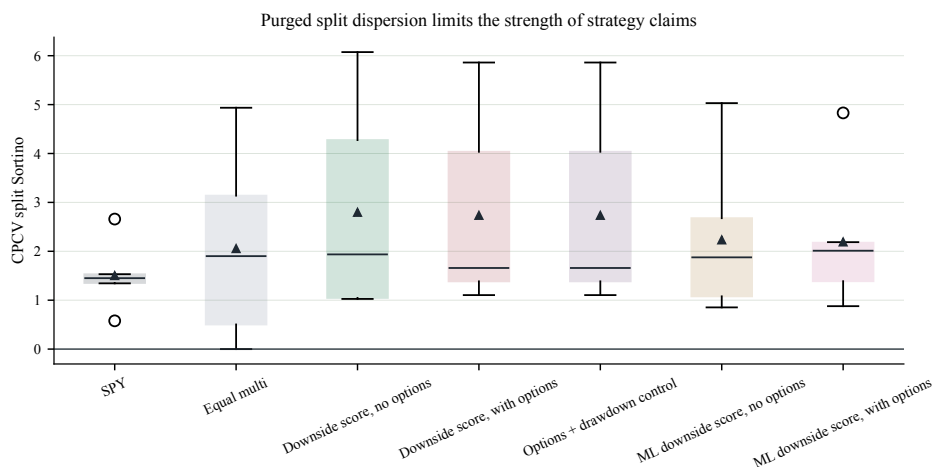


Figure 3: CPCV Sortino distribution across held-out folds. The figure shows how much rank uncertainty remains after purging and embargo, which is why the empirical claim is bounded to framework usefulness rather than deployable alpha.

Machine learning is evaluated under the same promotion standard. A forecast model must respect feature availability dates, purging, embargo, fold dispersion, and cost stress. It is then passed to the allocator as an uncertain expected-return input. If the portfolio-level result is worse than the simpler downside-score book, the ML arm remains a documented research result rather than a headline strategy.

6. Results and Discussion

The results section now answers the admission question directly. The strongest evidence admits the no-option cross-asset downside book. The option evidence is qualified because the option-enabled book is more defensive but gives up return. The ML evidence is kept visible as negative evidence because forecast validation does not translate into a better portfolio-level result.

6.1. Admission decision

Sleeve or rule	Decision	Main evidence	Caveat
Cross-asset downside book	Admitted	Return 15.12%, Sortino 2.00, max DD -19.87%.	Headline result is sample-bound, not live alpha.
Option sleeves inside allocator	Qualified	Max DD improves to -15.18% and CVaR95 to 4.63%.	Return falls to 13.07% versus the no-option book.
Fixed SPY put and collar overlays	Diagnostic	Benchmark max DD moves from -23.93% to -23.12% and -23.31%.	Not enough evidence for a standing overlay rule.
ML forecast arm	Rejected	ML option book Sortino 1.53 versus 2.00 for the simpler rule.	Forecast signal remains a research input.

Table 4: Admission decision summary. The table states what receives risk budget, what is qualified, and what remains diagnostic after costs, cashflows, and downside validation.

The table is the paper’s main result. It separates three ideas that are often blurred in multi-asset backtests. Diversification earns admission only when it improves the whole book after frictions. Convexity earns a qualification, not a blank check, because its premium must be repaid in the drawdown states that matter. Forecasting earns no capital allocation until its portfolio-level result beats the simpler rule.

6.2. Realized path

Strategy	Ann. ret.	Sortino	Max DD	CVaR95	Read
SPY	14.23%	1.14	-23.93%	9.51%	Base
60/40 SPY/TLT	8.70%	0.80	-25.82%	6.90%	Weak
Rolling inverse vol	12.48%	1.73	-19.36%	5.12%	Divers.
Downside score, no options	15.12%	2.00	-19.87%	5.66%	Admit
Downside score, with options	13.07%	1.99	-15.18%	4.63%	Qual.
ML downside score, with options	12.82%	1.53	-20.41%	6.64%	Research

Table 5: Headline out-of-sample evidence from February 2018 through May 2026. The table keeps only the metrics needed for the admission decision: return, downside reward, maximum drawdown, tail loss, and the resulting interpretation.

The no-option downside-score book is the strongest realized headline strategy. It does not win because more asset labels were added; it wins because the allocation rule uses the sleeves that improve the downside objective after costs. The option-enabled book has a different role. It reduces drawdown and tail loss, but the return give-up keeps it from replacing the no-option book as the main allocation.

Strategy	Equity	Commod.	Vol.	Rates	Options	Turnover
Rolling inverse vol	35%	38%	4%	23%	0%	3.0%
Downside score, no options	55%	27%	6%	12%	0%	15.0%
Downside score, with options	47%	23%	5%	9%	16%	15.7%
ML downside score, with options	46%	22%	2%	17%	13%	36.1%

Table 6: Average exposure buckets for the headline allocation rules. The table compresses individual sleeve weights into economic buckets so the reader can see what the admitted rules actually own without scanning every instrument.

6.3. Validation gates

Decision area	Read	Validation evidence	Limit
Cross-asset downside book	Supported	CPCV median Sortino 1.94 versus 1.45 for SPY.	Still sample-bound and not a live alpha claim.
Option-enabled book	Qualified	Block-bootstrap median max DD -12.71% versus -14.22% without options.	Realized return 13.07% versus 15.12% without options.
Cost stress	Stress pass	At 3x costs, no-option Sortino 2.52; option book 2.54.	Cost model remains a research approximation.
Backtest selection	Bounded	PBO 0.44 across 70 trials.	Validation reduces but does not remove selection risk.

Table 7: Compact validation summary. Each row reports how a validation screen changes the strength of the admission decision rather than listing every diagnostic output.

The validation layer makes the result useful but bounded. CPCV supports the broad ranking, while PBO and simulation remind the reader that the realized sample is short. Cost stress does not overturn the headline ranking, but it does not turn the study into a live trading claim. The point is to admit evidence at the strength it deserves.

Evidence block	Decision	Observed evidence	Interpretation
Options inside allocator	Qualified	Max DD -15.18% and CVaR95 4.63%.	Use as state-contingent defense, not return enhancement.
Fixed SPY put/collar overlays	Diagnostic	Put/collar max DD -23.12% and -23.31%.	Too close to SPY to promote as a standing rule.
Option state payoff	Conditional	Stress-state p95 payoff 2.68% for puts and 2.71% for collars.	Average payoff is not the admission test.
ML forecast arm	Rejected	Selected mean IC 0.499 without options and 0.481 with options.	Portfolio result does not beat the simpler downside rule.

Table 8: Qualified and negative evidence. Options, fixed overlays, option-state payoff, and ML forecasts are kept in the body only to show why their decisions are conditional or negative rather than hidden in the appendix.

This is the practical reading for a trader or portfolio manager. A sleeve enters the book only when it improves the whole portfolio after funding, spread, roll, premium, hedge, and drawdown costs. A sleeve is rejected when its apparent edge depends on stale marks, free rebalancing, unpriced roll, or a forecast that fails at the portfolio level. The body result is therefore not a menu of methods. It is an audit trail for the capital decision.

7. Conclusions

A sleeve is not an asset label. It is a funded cashflow with marks, frictions, hedge rules, and state-dependent payoffs. A SPY-centered investor should therefore admit additional equities, commodities, VIX futures, Treasuries, or SPY options only after the cashflow improves the portfolio objective that matters. In this study the objective is downside-aware performance after execution costs, not a clean in-sample frontier.

The admission audit produces a narrow result. The no-option cross-asset downside book is the headline admitted allocation in this sample. It improves the tested SPY path on return, Sortino, drawdown, and tail-risk evidence relative to the benchmark and simple baselines. The option-enabled book is qualified: it is more defensive and can improve drawdown and CVaR, but it gives up return. The machine-learning arm is rejected for headline allocation because forecast evidence does not become a better portfolio than the simpler downside rule.

The limitations are material. The analysis is not a live trading claim, the sample is limited, and several execution layers are simplified. Option execution uses quote anchors and model marks when daily matched quotes are unavailable; rates and futures execution use research proxies rather than brokerage-level order data; public and reproducible data constraints limit the granularity of constituent, roll, spread, and borrow assumptions. Production deployment would require richer market data, live brokerage cost modeling, point-in-time constituent reconstruction, executable futures and VIX-futures rolls, observed OPRA bid/ask execution, delta-hedged option sleeves, Treasury-futures cheapest-to-deliver accounting, and larger purged forecast-validation experiments.

The operational lesson is direct. Build the cashflow. Charge the friction. Choose the downside objective before fitting the allocator. Convexity is valuable when it pays in the drawdown states that matter, not when it looks attractive in isolation. Negative evidence belongs in the audit trail, not outside the paper. What remains is a portfolio construction method: admit instruments by state-contingent contribution and reserve capital-allocation claims for results that survive executable data and live trading constraints.

Data availability statement

The replication package contains the processed research panels and scripts needed to rebuild the reported tables and figures. External data feeds, when refreshed, are accessed through the documented build scripts and availability rules.

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A. Appendix

Reproducibility Commands

The core empirical run, manuscript audit, and test suite can be reproduced with:

```
python -m xasset_portfolio.run --config configs/base.yml
python scripts/paper_qa.py
python -m pytest
```

Appendix A is a supplement to the main empirical argument. It first shows additional realized-path and optimizer diagnostics, then checks option marks, hedge accounting, execution costs, futures rolls, and drawdown-control records. The validation section keeps simulation, stress, PBO, and claim-audit evidence together. Instrument conventions and solver notes are kept last because they support reproducibility and interpretation rather than the headline empirical claim.

A.1. Additional realized-path and optimizer results

This supplement begins with the realized-path displays because they are the quickest way to see what the model changes in practice. The figures use the same February 2018 to May 2026 out-of-sample panel as the body tables. They compare allocator behavior rather than re-estimating a separate experiment.

Strategy	Ann. ret.	Ann. vol.	Sortino	Sharpe	Max DD	Calmar	CVaR95	Hit rate
SPY	14.23%	16.56%	1.14	0.73	-23.93%	0.59	9.51%	67.0%
SPY + 10% put	14.49%	16.56%	1.18	0.75	-23.12%	0.63	9.32%	67.0%
SPY + 10% collar	14.47%	16.55%	1.17	0.75	-23.31%	0.62	9.34%	67.0%
60/40 SPY/TLT	8.70%	12.31%	0.80	0.53	-25.82%	0.34	6.90%	66.0%
Equal multi-asset	10.70%	13.19%	1.46	0.64	-27.80%	0.38	4.04%	59.0%
Rolling inverse vol	12.48%	9.71%	1.73	1.00	-19.36%	0.64	5.12%	65.0%
Downside score, no options	15.12%	10.35%	2.00	1.18	-19.87%	0.76	5.66%	70.0%
Downside score, with options	13.07%	8.87%	1.99	1.15	-15.18%	0.86	4.63%	67.0%
Options + drawdown control	11.82%	9.73%	1.59	0.94	-18.04%	0.66	4.97%	65.0%
ML downside score, no options	13.55%	12.38%	1.41	0.89	-22.72%	0.60	7.22%	68.0%
ML downside score, with options	12.82%	10.88%	1.53	0.93	-20.41%	0.63	6.64%	69.0%

Table 9: Full out-of-sample performance diagnostics. The body table keeps only the admission metrics; this supplement preserves volatility, Sharpe, Calmar, hit-rate, and other supporting columns for the full strategy set.

Strategy	Average OOS weights	Option notional	Monthly turnover
Equal multi-asset	SPY:17%, EQ_BASKET:17%, GC_FUT:17%, HG_FUT:17%, VX_FRONT:17%, TLT:17%	0.0%	0.0%
ML downside score, no options	SPY:36%, EQ_BASKET:19%, GC_FUT:13%, HG_FUT:11%, VX_FRONT:1%, TLT:19%	0.0%	39.8%
ML downside score, with options	SPY:30%, EQ_BASKET:15%, GC_FUT:11%, HG_FUT:11%, VX_FRONT:2%, TLT:17%, SPY_95P:7%, SPY_95_105_COLLAR:7%	13.2%	36.1%
Rolling inverse vol	SPY:20%, EQ_BASKET:16%, GC_FUT:23%, HG_FUT:15%, VX_FRONT:4%, TLT:23%	0.0%	3.0%
60/40 SPY/TLT	SPY:60%, TLT:40%	0.0%	0.0%
Downside score, no options	SPY:24%, EQ_BASKET:31%, GC_FUT:18%, HG_FUT:9%, VX_FRONT:6%, TLT:12%	0.0%	15.0%
Downside score, with options	SPY:18%, EQ_BASKET:29%, GC_FUT:14%, HG_FUT:9%, VX_FRONT:5%, TLT:9%, SPY_95P:7%, SPY_95_105_COLLAR:9%	15.8%	15.7%
Options + drawdown control	SPY:18%, EQ_BASKET:29%, GC_FUT:14%, HG_FUT:9%, VX_FRONT:5%, TLT:9%, SPY_95P:7%, SPY_95_105_COLLAR:9%	15.8%	15.7%
SPY	SPY:100%	0.0%	0.0%
SPY + 10% collar	SPY:100%, SPY_95_105_COLLAR:10%	10.0%	20.0%
SPY + 10% put	SPY:100%, SPY_95P:10%	10.0%	10.0%

Table 10: Full average out-of-sample weights. The body compresses weights into economic buckets; this table reports the instrument-level sleeve averages, option notional, and monthly turnover for all reported allocation rules.

Method	Use when	Current implementation	Main failure
Equal multi	alpha is weak	equal weights in six core sleeves	ignores expected return and state payoff
Inverse vol	risk estimates are usable but means are weak	rolling 36-month inverse-vol weights	can overweight low-vol drawdown assets
Constrained robust MVO	forecasts exist but estimation error is material	shrunk covariance, caps, L2/risk-budget penalties, costs	coarse grid for a lightweight reproduction
Downside score	Sortino/drawdown path is the mandate	candidate search with costs, caps, turnover	heuristic, not global proof
ML downside score	lagged forecasts survive purged CV	forecast-to-weight admission with frozen model	fold IC may not translate to portfolio value
CVaR/CDaR program	tail or path loss is explicitly capped	mathematical reduction in appendix	heavier solver and scenario-design burden

Table 11: Supplemental optimizer menu. The body states the admission rule; this table records the allocation rules used as diagnostics or candidate generators and the failure mode that prevents automatic promotion.

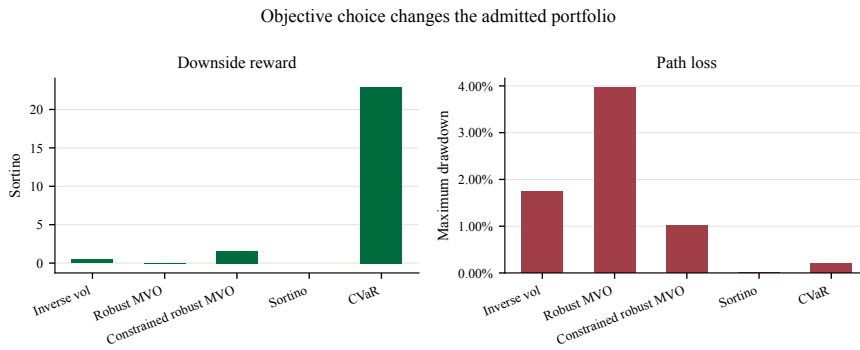


Figure 4: Objective comparison over the realized out-of-sample panel. The display compares the candidate allocation rules on the same net-return series, so differences reflect optimizer choice after common costs and instrument conventions rather than different input calendars.

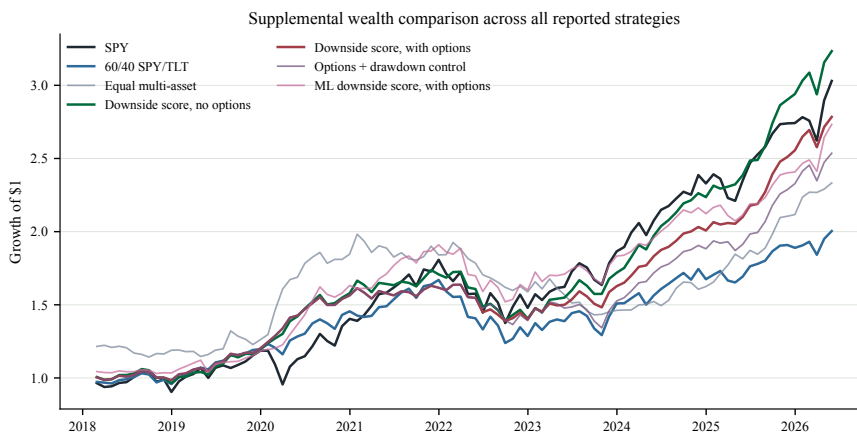


Figure 5: Supplemental wealth comparison for all reported strategies. The body figure keeps the headline comparison sparse; this appendix display shows the additional allocator and ML-assisted paths so the reader can see which alternatives were not promoted.

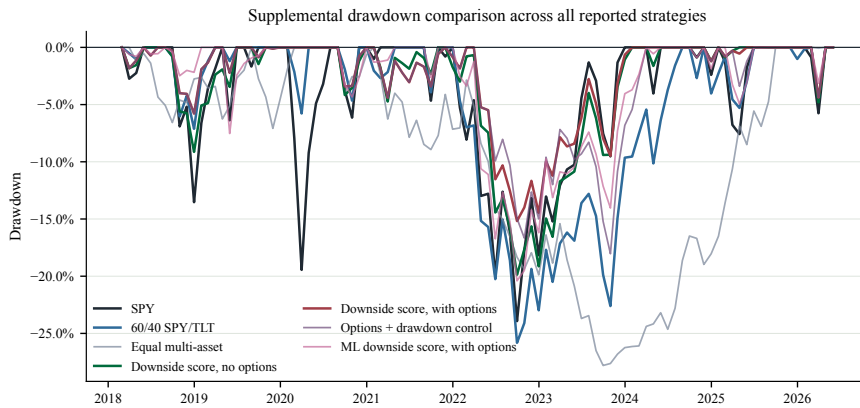


Figure 6: Supplemental drawdown comparison for all reported strategies. The figure keeps negative evidence visible by showing whether rejected or qualified strategies improved drawdown depth, duration, or recovery relative to the promoted downside-score books.

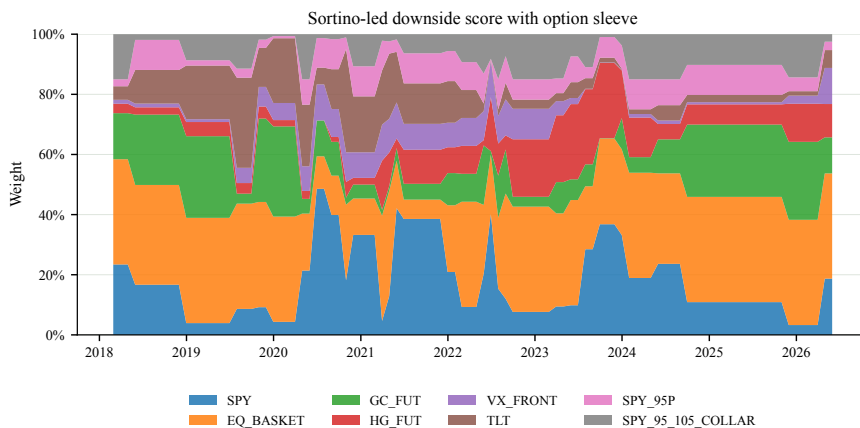


Figure 7: Sortino-allocation weights through time for the realized test window. The panels show how the downside-score allocator moves capital across SPY, diversifiers, and option sleeves as the rolling return and shortfall estimates change.

The compact body tables summarize the decision. The full displays above preserve the uncompressed strategy set for readers who want to audit the exact alternatives.

A.2. Option, hedge, and execution diagnostics

The option results are reported as diagnostics before they are treated as investment evidence. The tables below ask whether option marks, hedge accounting, execution costs, futures rolls, and drawdown-control trades are visible in the realized P&L. Each table earns its place by checking a specific way the backtest could otherwise look too clean: stale marks, free hedging, missing transaction costs, unpriced roll, or an overlay that reduces drawdown only by consuming hidden return.

Option sleeve	Mark source	Rows
SPY_95P	expiry_terminal_value	127
SPY_95P	model_mark_from_entry_iv	1537
SPY_95_105_COLLAR	expiry_terminal_value	127
SPY_95_105_COLLAR	model_mark_from_entry_iv	1537

Table 12: Option mark availability for the SPY put and collar overlays. Rows distinguish terminal expiry marks from model marks carried through the holding period, which is the minimum evidence needed before option sleeves can enter the optimizer.

Strategy	Hedge rows	Hedge trade cost	Hedge P&L
Downside score, with options	1324	0.08%	7.93%
Options + drawdown control	1323	0.08%	7.43%

Table 13: Delta-hedge reconciliation for option-admitting strategies. The table reports hedge rows, hedge trading cost, and hedge P&L over the realized execution panel, so the convexity result is not read as a free continuous-rebalancing result.

Strategy	Trades	Avg. turnover	Total cost
Downside score, no options	376	6.5%	0.93%
Downside score, with options	336	7.2%	2.67%
Options + drawdown control	345	11.9%	3.62%

Table 14: Execution-cost diagnostics for the daily panel. The comparison reports the costs charged to candidate strategies before monthly aggregation, keeping turnover and friction assumptions visible alongside realized performance.

Date	VX contract	Settlement	Rule
2013-01-02	VXF13	2013-01-16	nearest_settlement_delivery_avoidance
2013-01-15	VXG13	2013-02-13	nearest_settlement_delivery_avoidance
2013-02-12	VXH13	2013-03-20	nearest_settlement_delivery_avoidance
2013-03-19	VXJ13	2013-04-17	nearest_settlement_delivery_avoidance
2013-04-16	VXK13	2013-05-22	nearest_settlement_delivery_avoidance
2013-05-21	VXM13	2013-06-19	nearest_settlement_delivery_avoidance
2013-06-18	VXN13	2013-07-17	nearest_settlement_delivery_avoidance
2013-07-16	VXQ13	2013-08-21	nearest_settlement_delivery_avoidance

Table 15: Futures roll diagnostics for contracts used as portfolio sleeves. The table checks that roll effects are represented separately from spot exposure and are charged before futures compete with cash equities or ETFs.

Strategy	Triggers	Worst prior DD	Threshold
Options + drawdown control	22	-10.84%	-10.00%

Table 16: Drawdown-control overlay summary. The display compares the option-control policy against the base allocation on the same path and reports where the overlay reduces path risk versus where it consumes return.

A.3. Validation and simulation additional results

The validation appendix keeps negative and uncertain evidence in view. The simulation families are conditional uncertainty screens calibrated to realized strategy returns; they are not structural forecasts of the economy. The purpose is to keep the empirical result bounded. A strategy that looks attractive on the realized path still has to survive path resampling, drawdown-breach checks, stress windows, overfitting diagnostics, and the claim audit.

Strategy	Months	Legacy total	Daily-executed total	Mean diff.
SPY	100	202.69%	203.00%	0.00%
Downside score, no options	100	193.01%	223.36%	0.10%
Downside score, with options	100	143.35%	178.40%	0.14%
Options + drawdown control	100	143.35%	153.62%	0.05%

Table 17: Monthly proxy returns compared with monthly returns aggregated from daily execution. The reconciliation shows how much the execution layer changes the path before the monthly out-of-sample metrics are reported.

Strategy	Sortino p05	p50	p95	Median max DD	Positive
Equal multi	0.10	1.90	4.50	-15.73%	83.3%
ML downside score, with options	0.96	2.01	4.17	-8.45%	100.0%
Downside score, no options	1.03	1.94	5.75	-10.92%	100.0%
Downside score, with options	1.17	1.66	5.58	-9.83%	100.0%
Options + drawdown control	1.17	1.66	5.58	-9.83%	100.0%
SPY	0.77	1.45	2.38	-16.49%	100.0%

Table 18: Combinatorial purged cross-validation summary. Strategies are refit on purged training folds and evaluated on alternative held-out histories, so the table tests whether the realized ranking depends too heavily on one calendar path.

Strategy	Lag-1	Skew	Ex. kurt.	Recommended simulation
ML downside score, with options	-0.06	0.87	22.77	block bootstrap or egarch
Downside score, no options	-0.02	0.35	9.08	iid empirical resample ok
Downside score, with options	-0.03	0.63	15.06	iid empirical resample ok
Options + drawdown control	-0.02	0.09	8.49	iid empirical resample ok
SPY	-0.13	-0.29	13.07	block bootstrap or egarch

Table 19: Simulation diagnostics by strategy. Lag-1 autocorrelation, skewness, and excess kurtosis determine whether the supplemental paths should emphasize block bootstrap, EGARCH-style volatility clustering, or the weaker IID empirical baseline.

Strategy	Simulation	Sortino p50	Max DD p50	Terminal p05	Terminal p50
ML downside score, with options	block	1.67	-15.29%	1.69	2.71
ML downside score, with options	egarch	1.39	-21.67%	1.62	2.45
Downside score, no options	block	2.14	-14.22%	1.91	3.44
Downside score, no options	egarch	1.71	-20.31%	1.41	3.18
Downside score, with options	block	2.02	-12.71%	1.76	2.88
Downside score, with options	egarch	1.83	-16.66%	1.63	2.77
Options + drawdown control	block	1.82	-14.27%	1.57	2.57
Options + drawdown control	egarch	1.64	-18.39%	1.69	2.52
SPY	block	1.19	-30.60%	1.27	3.19
SPY	egarch	0.80	-43.23%	0.58	2.27

Table 20: Monte Carlo and bootstrap summary metrics. The table reports return, downside, tail, and wealth diagnostics across the supplemental path families for the same strategy set used in the body.

Strategy	Simulation	DD > 8%	DD > 10%	DD > 15%
ML downside score, with options	block	100.0%	100.0%	53.8%
ML downside score, with options	egarch	100.0%	98.3%	88.3%
Downside score, no options	block	100.0%	92.5%	43.8%
Downside score, no options	egarch	100.0%	98.3%	86.7%
Downside score, with options	block	100.0%	90.0%	23.8%
Downside score, with options	egarch	100.0%	95.0%	61.7%
Options + drawdown control	block	100.0%	93.8%	42.5%
Options + drawdown control	egarch	100.0%	98.3%	68.3%
SPY	block	100.0%	100.0%	98.8%
SPY	egarch	100.0%	100.0%	100.0%

Table 21: Drawdown breach probabilities in the supplemental simulations. The rows show how often each strategy crosses specified drawdown thresholds, which is the practical test behind the downside objective.

Regime	Strategy	Months	Ann. ret.	Max DD	Hit rate
calm	SPY	41	32.21%	-4.66%	78.0%
calm	Downside score, no options	41	28.79%	-4.17%	80.5%
calm	Downside score, with options	41	25.10%	-3.41%	82.9%
calm	Options + drawdown control	41	24.76%	-3.41%	82.9%
calm	ML downside score, with options	41	20.06%	-3.22%	80.5%
stress	SPY	59	3.19%	-25.58%	59.3%
stress	Downside score, no options	59	6.49%	-21.37%	62.7%
stress	Downside score, with options	59	5.40%	-18.61%	55.9%
stress	Options + drawdown control	59	3.62%	-25.15%	52.5%
stress	ML downside score, with options	59	8.05%	-21.84%	61.0%

Table 22: Regime-conditional returns. Regimes are descriptive cuts rather than optimizer inputs, so the table is an interpretation aid for where each allocation earns or loses money.

State	Overlay	Months	Mean	P95 payoff	Positive
calm	SPY 95P	68	-0.23%	-0.08%	0.0%
calm	SPY 95 105 COLLAR	68	-0.36%	0.01%	5.9%
stress	SPY 95P	68	0.09%	2.68%	10.3%
stress	SPY 95 105 COLLAR	68	0.05%	2.71%	10.3%

Table 23: Option overlay payoffs by state bucket. The table checks whether option premium is repaid in the portfolio states where convexity is needed, rather than only on average over the full sample.

Strategy	Costs	Ann. ret.	Sortino	Max DD
Downside score, no options	1x	15.12%	2.55	-19.87%
Downside score, no options	3x	14.98%	2.52	-19.96%
Downside score, with options	1x	13.07%	2.66	-15.18%
Downside score, with options	3x	12.61%	2.54	-15.36%
Options + drawdown control	1x	11.82%	2.19	-18.04%
Options + drawdown control	3x	11.35%	2.09	-18.58%
ML downside score, with options	1x	12.82%	2.01	-20.41%
ML downside score, with options	3x	11.95%	1.86	-20.88%

Table 24: Cost multiplier stress. Strategies that work only under optimistic frictions are kept out of the strongest claim set, so this table preserves the cost-sensitive diagnostic detail behind the compact body summary.

Window	SPY	60/40	Equal multi	Downside score no options	Downside score with options
COVID shock	-9.18%	1.87%	28.94%	12.13%	13.32%
2022 inflation/rates	-17.74%	-24.09%	-13.18%	-16.22%	-12.86%
2025-26 late sample	30.08%	19.67%	43.58%	44.63%	38.68%
Full OOS	203.00%	100.42%	133.29%	223.36%	178.40%

Table 25: Stress-window return diagnostics. The table compares strategy returns in identified stress windows and keeps the hedge question tied to the parts of the path that matter for peak-to-trough loss.

Status	PBO	Trials	Metric
ok	0.44	70	sortino

Table 26: Probability-of-backtest-overfitting diagnostic. The table summarizes whether the best in-sample configuration tends to retain rank out of sample across the cross-validation partitions.

Strategy	Model	Selected	Mean IC	IC p05	IC p95	Folds
ML downside score, no options	ridge alpha 1	no	0.497	0.337	0.625	5
ML downside score, no options	ridge alpha 5	yes	0.499	0.339	0.598	5
ML downside score, no options	lgbm leaves 7	no	0.478	0.355	0.570	5
ML downside score, no options	lgbm leaves 15	no	0.458	0.348	0.552	5
ML downside score, with options	ridge alpha 1	no	0.473	0.415	0.539	5
ML downside score, with options	ridge alpha 5	yes	0.481	0.419	0.562	5
ML downside score, with options	lgbm leaves 7	no	0.433	0.211	0.556	5
ML downside score, with options	lgbm leaves 15	no	0.446	0.220	0.576	5

Table 27: Purged-CV forecast validation. The table reports model-level forecast evidence before the forecasts are allowed to affect a portfolio-level allocation.

Claim	Type	Status
Optimizer decision tree	standard	source anchored
Downside/CVaR/drawdown objectives	standard	source anchored
Cashflow normalization	original framing	proved/sketched here
ML forecast band	adapted	validation caveat
Objective comparison	empirical	run manifest

Table 28: Claim-audit summary for the manuscript. The entries separate source-grounded finance claims, adapted modeling choices, original framework language, and empirical claims produced by the replication workflow.

A.4. Instrument convention reference

Every candidate sleeve is first translated into domestic cashflows. An equity sleeve records shares, dividends, borrow cost, corporate actions, and FX conversion when the underlying is foreign. An ETF adds tracking error, fund expense, distribution treatment, and creation-redemption liquidity. A listed option records premium, exercise style, settlement, mark source, surface assumptions, Greeks, and the hedge policy used to convert convex payoff into portfolio cashflows.

Futures are variation-margin instruments, not cash purchases of the underlying. The convention file records multiplier, expiry, roll rule, margin drag, collateral return, and delivery rule when relevant. Treasury futures additionally require a deliverable basket, conversion-factor, cheapest-to-deliver, financing, and basis treatment in a production implementation. FX and forwards record spot conversion, domestic and foreign cash rates, forward points, settlement form, basis, and roll spread. Fixed-income sleeves are represented as coupon strips with level, slope, curvature, key-rate DV01, spread duration, convexity, financing, and inflation beta where relevant.

Instrument	Required convention	Main exposure	Common research error
Equity	Shares, dividend, borrow, FX if foreign	Spot beta and factor beta	Treating correlated equities as diversifiers
ETF	NAV, fee, tracking, distribution, creation basket	Underlying basket and tracking error	Ignoring fund structure and liquidity
Option	Premium, exercise, settlement, vol surface, hedge policy	Delta, gamma, vega, theta	Treating delta hedging as free
Future	Multiplier, margin, expiry, roll, delivery rule	Spot, curve, roll, collateral	Treating notional like cash equity
Forward/FX	Maturity exchange, discounting, basis, settlement	Carry, basis, spot FX	Mixing pricing drift with alpha
Bond	Coupon strip, day count, curve, financing	DV01, key-rate DV01, convexity	Using one duration for curve trades

Table 29: Instrument convention checks before a sleeve enters the allocator. The table shows what must be specified, which exposure is being bought, and the common shortcut that would make the backtest look cleaner than the traded instrument.

A weak data-quality score does not automatically exclude an instrument. It widens the forecast haircut and raises the admission hurdle. Examples include option quotes that survive mid-price checks but fail open-interest filters, futures returns that disappear after executable roll windows, or an FX carry sleeve that fails after funding basis.

A.5. Solver reductions, proof sketches, and implementation notes

The final appendix section records the mathematical reductions and implementation interfaces needed to reproduce the study. The body keeps the economic object in prose; the details below explain how the same cashflow cube is scored under alternative objectives.

Let $x \in \mathbb{R}^n$ denote sleeve weights, $r_s \in \mathbb{R}^n$ scenario net instrument returns, and p_s scenario probabilities. The net portfolio return is

$$R_s(x) = r_s^\top x - C_s(x),$$

where C_s contains spreads, commissions, impact, funding, margin drag, roll cost, hedge cost, and data-quality penalties. The same scenario cube feeds each optimizer; only the objective and admission rule change.

The robust mean-variance screen starts from

$$\max_{x \in \mathcal{X}} \hat{\mu}^\top x - \rho \|A^\top x\|_2 - \frac{\lambda}{2} x^\top \hat{\Sigma} x - C(x).$$

The cookbook-inspired implementation uses identity-target covariance shrinkage,

$$\hat{\Sigma}_{\text{shr}} = (1 - \alpha) \hat{\Sigma} + \alpha \frac{\text{tr}(\hat{\Sigma})}{n} I,$$

and scores deterministic candidate portfolios after cost, prior, and risk-budget penalties. This preserves the economic logic of robust optimization without requiring a heavy solver in the public reproduction path.

For target return τ , the Sortino objective is

$$S_\tau(x) = \frac{\sum_s p_s R_s(x) - \tau}{(\sum_s p_s (\tau - R_s(x))_+^2)^{1/2}}.$$

The denominator is a lower partial moment, not total volatility. A sleeve enters only when its robust mean contribution clears the shortfall-risk and cost hurdle. The CVaR loss program uses the Rockafellar-Uryasev representation,

$$\text{CVaR}_\alpha(L(x)) = \min_{q \in \mathbb{R}} \left[q + \frac{1}{1 - \alpha} \sum_s p_s (L_s(x) - q)_+ \right],$$

which is useful when the mandate cares about expected loss beyond a quantile.

Drawdown is path-specific. With cumulative wealth $W_t(x)$, running peak $P_t(x)$, and drawdown $D_t(x) = P_t(x) - W_t(x)$, a hedge can be admitted even with negative standalone mean when it pays in the states that set the running peak-to-trough loss. This is why the paper evaluates option overlays in realized drawdown states instead of judging them only by unconditional average return.

The implementation interface is intentionally small. The package exposes `InstrumentSpec`, `FrictionProfile`, `ObjectiveSpec`, `ForecastSpec`, `ValidationSpec`, `ScenarioCube`, `HedgePolicy`, and `BacktestResult`. The reproduction command

```
python -m xasset_portfolio.run --config configs/base.yml
```

loads processed panels, builds the scenario cube, evaluates the allocator set, writes the reported tables and figures, and records the run manifest. Larger promoted studies can add data feeds or production instrument conventions behind the same interfaces without changing the manuscript contract.